



MapBiomás Argentina Fuego

ATBD

Algorithm Theoretical Basis Document

Collection 1

Version 1.0

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1 Introduction

1.1 Scope and purpose of this document

This document describes the theoretical basis, the methodological framework and the operational considerations behind Collection 1 of MapBiomass Argentina Fuego: an annual, 30 m burned-area product covering the whole of Argentina for the period 1999–2025. It succeeds the ATBD of the Patagonian pilot (Collection 0), whose structure and concepts it inherits, and whose algorithm it replaces almost entirely.

This ATBD is conceptual, and it is deliberately not a complete specification of the algorithm. Every stage described below carries dozens of decisions — thresholds, window lengths, encodings, tiling schemes, masks, edge cases — and writing all of them down here would produce a document several times this length that nobody would read. What follows is the *reasoning*: what each stage measures, why it measures it that way, and what it hands to the next stage. Readers who want to reproduce the algorithm elsewhere, or who need the exact value of a parameter, should go to the per-step technical notes in the project repository, at collection-01/docs (<https://github.com/barberaivan/mapbiomas-argentina-fire>). There is one note per workflow step, numbered to match it, and each one points onward to the production code, the configuration files and the analysis notebooks that justify its choices. Where this document says “see docs/NN”, that is the file it means.

Validation and the statistical characterisation of the product are outside the scope of this document. They are separate activities that *consume* the finished map rather than being stages of it, and they are documented on their own.

1.2 Objectives of Collection 1

Collection 0 had two goals: to assemble a working group capable of producing burned-area maps for MapBiomass Argentina, and to develop a mapping algorithm of reasonable accuracy with a relatively low demand of human effort for training-data collection. Both were met at the scale of one region. Collection 1 sets out to

- extend the mapping to the **entire national territory** and to a multi-decadal Landsat record;
- replace the parts of the pilot algorithm that did not generalise — in particular the hand-drawn masks and the empirical object filter — with **data-driven, reproducible components**; and
- deliver the set of burned-area subproducts that the MapBiomass Fuego network publishes for every participating country (Alencar et al. 2022).

1.3 Spatial and temporal extent

Collection 1 maps the whole of Argentina, buffered by approximately 2 km so that no coastal or border pixel is clipped, and works over the five regions defined by MapBiomass Argentina — Patagonia; Monte, Puna and Altos Andes (“Cuyo”); Chaco; Pampas; and Atlantic Forest (Fig. 1). The regions are not processing units in the sense that the pilot anticipated: they enter the algorithm as a *covariate* (Section 3.4), while the actual processing runs over the national 1:250,000 chart grid, the *cartas* (Section 2.2), or — from the object stage onwards — over the whole country at once.



Figure 1: The five regions defined by MapBiomass Argentina (black). The blue rectangle is the Collection 0 pilot area, shown here only for reference: Collection 1 maps the whole country.

The spectral stage evaluates Landsat observations from 1998 onwards and produces annual layers for the calendar years **1999–2025**. The year 1998 is used only as the previous-year reference that the annual metrics require. From the segmentation stage onwards the algorithm works in a **fire year** running from 1 May to 30 April (Section 5.2), which yields fire years 1998–2025, the first and the last being partial by construction.

1.4 Input datasets

Collection 1 was developed on complete Landsat time series plus two ancillary MapBiomass Argentina layers, processed almost entirely in Google Earth Engine (GEE; Gorelick et al. 2017).

1.4.1 Landsat imagery

All available Landsat Collection 2 Level-2 (surface reflectance) scenes at 30 m intersecting the territory and the period were used (U.S. Geological Survey 2021b). During the mapped period the active sensors were Landsat 5 TM (until 2013), Landsat 7 ETM+ (from mid-1999), Landsat 8 OLI (from 2013) and Landsat 9 OLI-2 (from 2021). The six reflective bands — blue, green, red, near infrared (NIR) and the two shortwave infrared bands (SWIR1, SWIR2) — are read, and a set of spectral indices is derived from them (Section 3.6).

Scenes acquired on the same date are collapsed into a single date-mosaic before anything else happens, so that the unit the algorithm reasons about is a **date**, not a scene footprint. A *carta* is covered by roughly 150 such dates in a padded focal year.

1.4.2 Quality filters

Landsat observations are masked with the QA_PIXEL band delivered with the Collection 2 surface reflectance products, excluding cloud, dilated cloud, cloud shadow, snow and water.

No temporal interpolation and no spatial gap-filling are applied. A missing observation stays missing. This is a deliberate constraint rather than an omission: the temporal stage (Section 4) reads real acquisition dates and counts real observations, and both would be corrupted by synthetic values. The direct consequence is that observation density varies in space and in time — strongly so before 2000, when only Landsat 5 and a newly launched Landsat 7 were flying — and the algorithm carries that density forward as an explicit quality channel rather than hiding it.

1.4.3 Spectral harmonisation between sensors

Because the series combines sensors with slightly different spectral responses, Landsat 5 TM and Landsat 7 ETM+ reflectances are transformed into the Landsat 8/9 OLI domain band by band, using the coefficients of Roy et al. (2016). OLI and OLI-2 observations are left untouched. This reduces artificial discontinuities at sensor transitions, which in a change detector would otherwise be read as change.

1.4.4 MapBiomass Argentina land cover

The annual land use and land cover (LULC) classification of MapBiomass Argentina (MapBiomass Argentina 2025) is used for two distinct purposes, and both read the **previous year** ($y - 1$) relative to the year being mapped:

1. it defines the **fire class** (`veg_fire`) of every pixel, which selects which burn-probability model judges that pixel, and which surfaces are excluded from mapping altogether as non-burnable (Section 3.4);
2. it supplies the object-level vegetation composition metrics used by the object classifier (Section 6.2).

Reading $y - 1$ is not a detail. The land cover of a burned pixel changes *because* it burned, so using the focal year's classification would let the fire we are trying to detect decide how it is detected. The same rule is applied everywhere the classification is consumed.

1.4.5 MapBiomass annual mosaic

The second ancillary layer is the MapBiomass Argentina **annual mosaic** — the reflectance summary that the land-cover initiative itself classifies. Forty of its bands are used: the six optical bands plus NDVI, NDWI, NPV and NDFI (Souza Jr., Roberts, and Cochrane 2005), each summarised as annual median, dry-season median, wet-season median and standard deviation. Again, the mosaic of the **previous** year is attached to every observation.

This layer is what gives the burn-probability model its *baseline*. A burn scar is not a spectral value; it is a departure from what that particular pixel normally looks like, and how large a departure counts depends on the pixel's productivity, seasonality and spectral variability. The previous-year mosaic encodes all three, and it does so per pixel, which is what allows a single model to work across an enormous productivity gradient. Collection 0 built the same idea with hand-made robust annual minima and maxima of four indices; Collection 1 replaces them with a layer that MapBiomass already computes, is already quality-controlled, and is far richer.

1.5 Computing platforms

Most of the pipeline runs in GEE, which stores the entire Landsat archive and the MapBiomass products and provides the compute to evaluate a model over them. Two stages run locally: the fitting of the burn-probability models, in R (R Core Team 2025) with `glmnet` (Friedman, Hastie, and Tibshirani 2010), and the entire object stage — vectorisation, object metrics and the object

classifier — also in R, on tabular and raster data downloaded from GEE. Only coefficients and object attributes cross between the two environments, never imagery.

The GEE compute project is shared with the whole MapBiomass Fuego network, which has one practical consequence worth recording: every task launched by Argentina carries a namespaced description, and no task is ever inspected or cancelled on the basis of a generic name.

2 The burned-area mapping algorithm

The algorithm detects burned area at the finest temporal and spatial resolution Landsat allows — one 30 m pixel on one observation date — by exploiting the **spectral**, **temporal** and **spatial** signature of fire, in that order. Each stage consumes what the previous one produced and adds a kind of evidence the previous one could not see (Fig. 2).

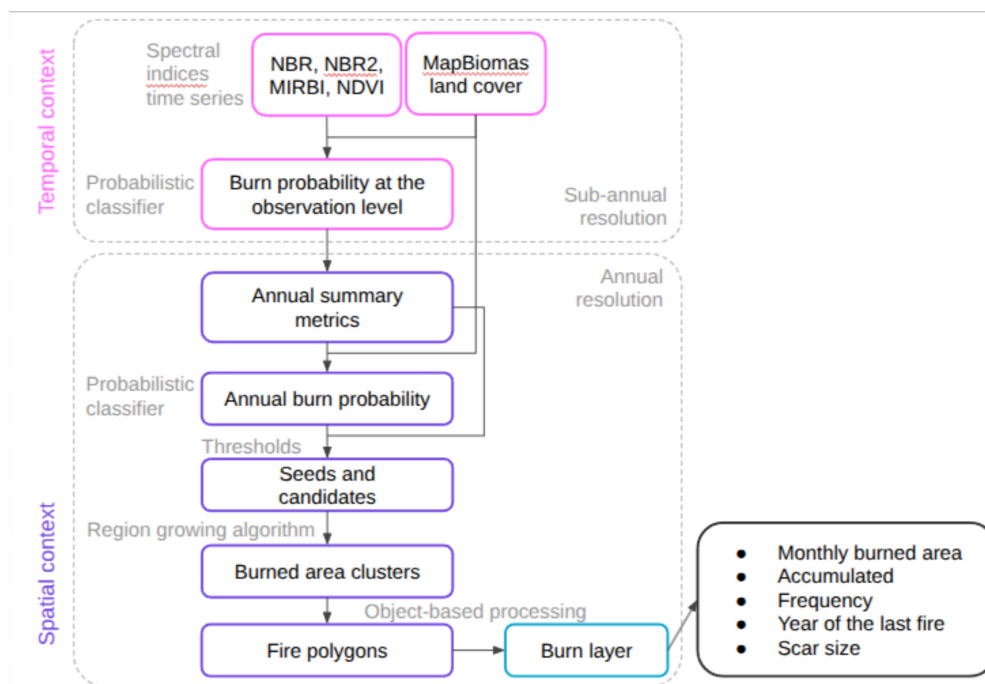


Figure 2: **Placeholder.** The diagram of the Collection 0 algorithm, shown here while the Collection 1 diagram is being prepared. The overall shape — spectral, then temporal, then spatial context, then the published subproducts — still holds, but several boxes have changed: Collection 1 has no annual probabilistic classifier and no hand-drawn masks, and its object filter is a fitted classifier rather than a set of thresholds.

Spectral — is this observation burned? A probabilistic classifier scores a single pixel on a single date, from its own reflectance and indices plus previous-year context. The output is a **burn probability per observation**. Nothing is decided here.

Temporal — was this pixel burned this year, and when? A burn probability on one date is not a detection: cloud shadow, wet soil and dark surfaces all raise it. What distinguishes fire is what the probability does *through time* — it rises and *stays* risen. The second stage quantifies, per pixel and per calendar year, a **maintained increase** in burn probability, together with the date on which it happened.

Spatial — is this pixel part of a fire? Fires are patches, not independent pixels. The third stage brings that context in twice: a region-growing algorithm converts per-pixel evidence into scar **objects**, and a classifier then judges each object on properties no pixel has — shape, size, vegetation composition, seed density, the spread of burn dates.

2.1 The unifying principle: keep quantities, decide late

One idea runs through all three stages: **quantitative information is preserved for as long as possible**, so that every later step can weigh the earlier evidence instead of inheriting someone else's verdict. The temporal stage analyses a burn probability, not a hard burned/unburned class. The region-growing step judges pixels on numeric metrics, which is the only reason the seed/candidate distinction can be expressed at all. The object classifier consumes continuous metrics and returns a *probability* of fire, which is used to filter but is also published, so a user can apply a stricter or a looser threshold than ours. A pipeline that thresholded early would be simpler, and would throw away exactly the evidence the next stage needs.

2.2 The processing grid

Every image-based step runs over the MapBiomass Argentina working grid — the national 1:250,000 chart series, locally called *cartas*. Each sheet covers roughly 14,000 km², and 248 of them intersect the buffered country. Tiling by chart sheet rather than by Landsat WRS-2 path/row is what lets one step read another step's tile and assemble the results on a single lattice with no resampling anywhere in the chain. All exports pin the projection and the affine transform explicitly, never a nominal scale, because in geographic coordinates a nominal 30 m scale is a *different* grid.

3 Spectral stage: burn probability per observation

3.1 Foundations

The unit of judgement is the observation, not the year. The model asks, for a single pixel on a single date, how likely it is that this pixel is burned, using spectral information only. Every usable Landsat observation is evaluated, so a fire is detected as a *change in the series* rather than as a signature in a summary image. The approach follows Long et al. (2019), where all Landsat observations are scored.

This is the opposite end of the design space from MapBiomass Fire Brazil, which classifies an **annual mosaic** — the same way land cover is mapped everywhere in the network. Classifying a mosaic is cheaper and reuses the land-cover machinery, but it collapses the time axis exactly where the information is: the date of the burn, the pre-fire baseline of that specific pixel, and the distinction between a scar and a spectrally similar surface that was always dark. Keeping the observation as the unit is what makes the temporal stage possible at all.

It is also a choice about *time granularity*. There are methods that avoid discretising time altogether: CCDC fits a continuous harmonic model per pixel and flags breaks in it (Z. Zhu and Woodcock 2014), and CODED extends that idea to forest degradation with spectral unmixing (Bullock, Woodcock, and Olofsson 2020). MapBiomass works in discrete annual units, and so does this collection: the product is one burned-area map per year, so aggregating the raw Landsat information by year is the natural shape for everything upstream of it.

The consequence for training data is direct: **a training point is not a label**. The label is per observation, assigned from where that observation's date falls relative to the fire.

3.2 Training data

Training points were collected interactively by domain experts, fire by fire. For each selected fire the collector delimits a pre-fire and a post-fire window on the Landsat record, and then places burned and unburned points inside the scar and its surroundings, aiming to cover as much of the spectral variability of each class as possible and inspecting index time series wherever the classification is doubtful.

Collection 1 extends this protocol from 30 fires in one region to **198 fires across the five regions** (13 in Atlantic Forest, 30 in Cuyo, 37 in Pampas, 47 in Patagonia, 71 in Chaco), holding **147,222**

points. The full Landsat time series is then extracted at every point, which turns those points into **6,177,098 observations**. Labels are assigned by window: observations of a burned point after the fire are labelled burned; observations of a burned point before the fire, and all observations of unburned points, are labelled unburned.

The efficiency argument made in the pilot ATBD holds and is worth restating, because it is the reason this protocol was kept: each collected point yields roughly forty labelled observations, so the sampling effort buys an order of magnitude more training data than placing one point per observation would. Observations from the same point are temporally correlated, so the effective gain is smaller than the nominal one — but the time series is needed anyway, since it is the series, not the single observation, that the rest of the algorithm reads.

Fires with no burned points are collected on purpose. Several of the sampled “fires” are in fact drought-affected steppe or volcanic-ash deposits — events that look like fire in the indices and were the largest source of commission error in the pilot. They enter the training set as unburned-only, and dropping them would remove the hardest negatives the model ever sees.

3.3 The cleaning gate

Collectors make mistakes: a post-fire window set too generously, a date that belongs to the next fire, an entire fire that should not be in the training set at all. A mandatory cleaning pass sits between the download of the observations and the fit, adding a boolean `fit` column that marks which observations the model is allowed to see; the fitting script refuses to run without it.

The pass has two layers. A **base hard filter** keeps only observations inside the valid window for the point’s type, from the collector’s own pre/post dates. On top of it sit **per-fire manual edits**, transcribed from a reviewed workbook — typically of the form “keep only the first three post-fire dates” or “drop this fire”. The edits are ranked over the fire’s *unique dates*, pooled across all of its burned points, not over each point’s own observation count; that distinction is the single most common way to mis-apply such a rule.

Of the 6,177,098 extracted observations, **5,923,062 pass the gate**, of which **541,789 (9.1 %) are labelled burned**. The edits are verified visually on per-fire panels that plot the fitted burn probability against the raw indices and mark held-out dates (Fig. 3).

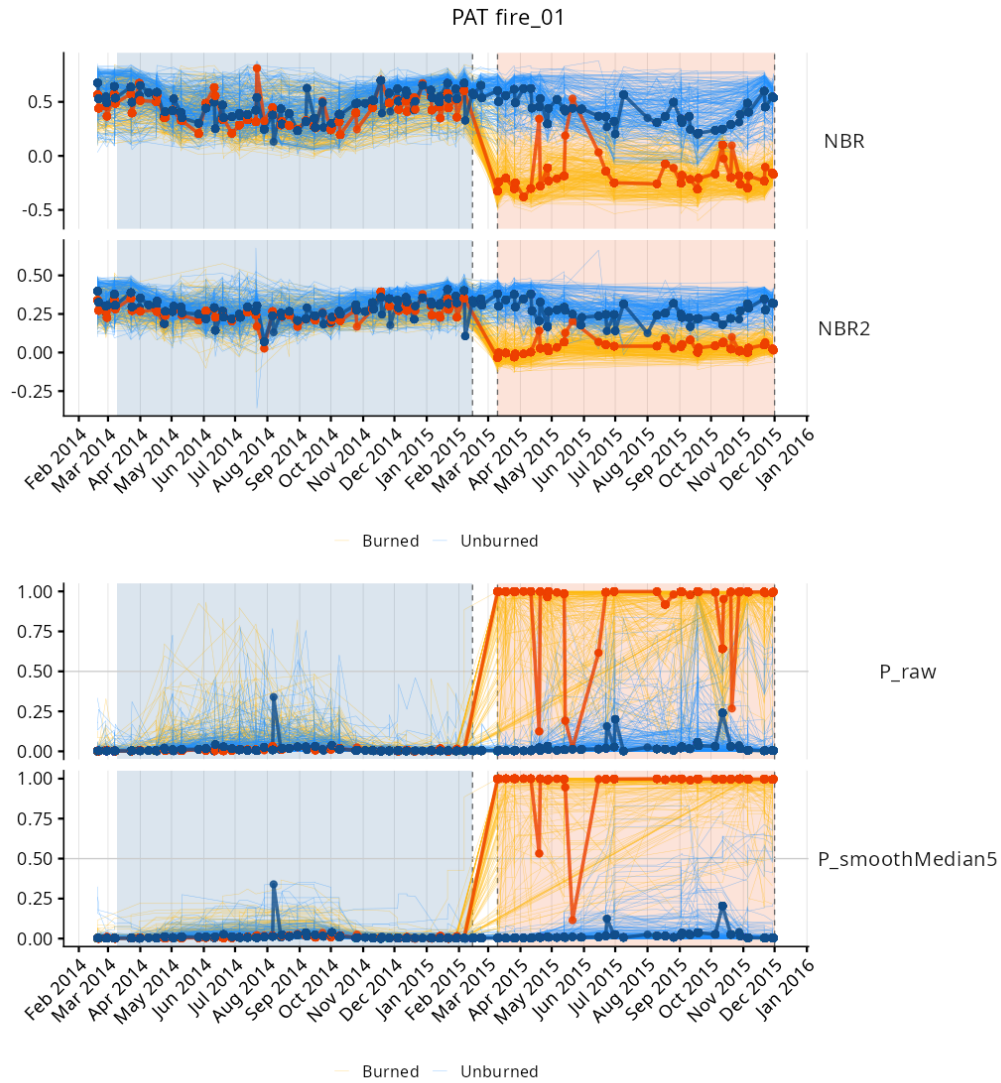


Figure 3: Diagnostic panel for one collected fire (Cholila, Patagonia, 2015). Thin lines are individual training points, thick lines the median across points; orange is the burned class, blue the unburned class. The top two rows show the raw NBR and NBR2 series and the shaded pre-fire and post-fire windows defined by the collector; the bottom two show the predicted burn probability per observation, raw and smoothed. Unburned points keep a low probability throughout, while burned points jump to near 1 at the fire and stay there — the signature the temporal stage is built to measure.

3.4 Fire classes: the veg_fire remap

A separate model is fitted per vegetation type, because burn signatures differ by fuel: what a scar does to the spectral series of a closed forest is not what it does in grassland or open shrubland, and one pooled model would fit the average of surfaces that do not behave alike.

The unit of stratification is the `veg_fire` class, obtained by crossing the MapBiomass region with the MapBiomass LULC class of the previous year and remapping the result through a single lookup table. The table yields **23 fittable classes** plus two non-fittable sentinels — *non-burnable* (water, rock, ice, urban, bare) and *non-observed*. Only the 23 are ever predicted, so the non-burnable mask that the network’s post-processing applies at the end is, in this algorithm, applied at the very beginning.

The class names carry both the vegetation type and the region it belongs to: `forest_pat`, `grassland_pampa`, `shrubland-open_chaco`, `agriculture_chaco`, and so on. **The grouping is not purely ecological — data availability was a hard constraint.** A strictly within-region remap was considered first, but several classes are better off crossing regions: partly for spectral and land-use similarity, partly because data collection was organised by region and that did not guarantee that every class would be represented in every region. So a class may span regions — `agriculture_cuyo-pat` covers both Cuyo and Patagonia. **The remap is an open path for improvement:** with more collected fires, groups currently merged for want of data could be split on their own merits.

Classes backed by few fires are fitted anyway rather than dropped or merged further; the alternative removes a vegetation type from the map entirely. The cost is visible in the cross-validation, where the number of folds is capped by the number of fires carrying positives.

3.5 Model structure

The model has to be deployable inside GEE, and that constraint chose it. A fitted logistic regression is a set of coefficients plus a simple equation: trivial to store, to version and to ship as a CSV, and cheap to evaluate over billions of pixel-dates. The richer classifiers GEE offers natively — random forest, boosted trees — fail at both ends here. It is doubtful they could be fitted on this many training observations, and a fitted one cannot be saved as an asset, which is precisely what deploying over the whole Landsat archive requires. External machine-learning services would lift both limits at a large step up in complexity and at a monetary cost to the initiative.

We also wanted a **natively probabilistic** model, because everything downstream consumes a probability rather than a hard class: the temporal metrics read the *shape* of p through time. The GEE constraint bites again — a probability-mode random forest is not exportable, and extracting calibrated probabilities from a regression-mode forest on binary data means tuning minimum node size against N , a dependency that would have to be re-tuned per class. Logistic regression gives the probability by construction, with nothing to tune for it.

The probability that the observation of pixel c at time t corresponds to burned is

$$\Pr(y_{c,t} = 1) = \frac{1}{1 + \exp(-\eta_{c,t})}, \quad (1)$$

$$\eta_{c,t} = \alpha_{k[c]} + \sum_{j=1}^J \beta_{j,k[c]} x_{j,c,t},$$

where α and β are estimated parameters, $k \in \{1, \dots, 23\}$ is the `veg_fire` class, and the notation $k[c]$ indicates the class of pixel c in the year *previous* to the one time t belongs to. $J = 51$ is the number of regression terms in the deployed model.

Every class shares one predictor set, chosen globally rather than per class: only the coefficients differ between vegetation types. That keeps deployment cheap in the same way the model itself does — the prediction pipeline builds one band set once and reuses it for all 23 classes, instead of assembling a different design per vegetation type and paying for it at every pixel-date.

3.6 Predictors

Twenty-one spectral features are computed for every Landsat observation: the six optical bands and fifteen indices — NBR (Key and Benson 2006), NBR2 (U.S. Geological Survey 2021a), MIRBI (Trigg and Flasse 2001), NDVI (Tucker 1979), the three tasselled-cap components (Baig et al. 2014), NDMI (Gao 1996), NDSI (Hall, Riggs, and Salomonson 1995), SAVI (Huete 1988), NDWI (McFeeters 1996), AFRI (Karnieli et al. 2001), kNDVI (Camps-Valls et al. 2021), EVI2 (Jiang et al. 2008) and NIRv (Badgley, Field, and Berry 2017). Forty previous-year mosaic bands are available alongside them.

Four fire-sensitive indices are worth writing out, since they are the ones the pilot used and the ones that still carry most of the fire signal:

$$\begin{aligned} \text{NBR} &= \frac{\text{NIR} - \text{SWIR2}}{\text{NIR} + \text{SWIR2}}, & \text{NBR2} &= \frac{\text{SWIR1} - \text{SWIR2}}{\text{SWIR1} + \text{SWIR2}}, \\ \text{NDVI} &= \frac{\text{NIR} - \text{RED}}{\text{NIR} + \text{RED}}, & \text{NDMI} &= \frac{\text{NIR} - \text{SWIR1}}{\text{NIR} + \text{SWIR1}}. \end{aligned}$$

The candidate design was much larger than the deployed one. It began as several hundred terms — all focal features, all mosaic summaries and a large set of interactions — and was reduced in two steps, first to 129 terms and then, by ranking those terms globally and cutting the ranking, to the deployed set. Predictive skill is flat from the full fit down to about fifty terms and falls below that, so the smallest set on the flat part was deployed. The reduction is not cosmetic: *the number of terms is the dominant cost of the prediction stage*, since each term becomes one image band that must be built and multiplied at every pixel-date. The deployed set costs about 25 % less compute per tile than the 129-term one.

The deployed model carries an intercept plus 51 terms, in five blocks:

- **10 focal main effects** — GREEN, RED, NIR, SWIR1, SWIR2, NBR, NBR2, NDVI, NDMI, NDSI, all measured on the observation being scored;
- **14 previous-year mosaic main effects** — medians, dry/wet composites and standard deviations of the optical bands and of NDVI, NDWI, NPV and NDFI;
- **10 focal × focal interactions** — e.g. NBR × NBR2, SWIR1 × NDVI;
- **4 same-band interactions** — a focal band with its own previous-year median (GREEN, NIR, SWIR2, NDVI), which is the model’s explicit *change* term: the product of today’s value with this pixel’s baseline;
- **13 cross interactions** between previous-year mosaic summaries and focal fire indices or focal bands.

The design deserves a word, because it is what makes a linear model work across a country. A logistic regression on raw indices alone would be exactly the “one threshold everywhere” approach that the pilot ATBD argued against; the optimal NBR cut for a Patagonian forest is not the optimal cut for Chaco open shrubland. The previous-year mosaic main effects let the model shift its intercept with the pixel’s normal appearance, and the cross and same-band interactions let it change the *slope* of its response to a fire index according to that appearance. Fitting a separate model per `veg_fire` class handles the between-type differences; the interactions with the previous-year baseline handle the within-type gradient in productivity, structure and seasonality.

3.7 Fitting, regularisation and cross-validation

The design is strongly collinear by construction — indices share bands, interactions share factors — so the models are fitted by **elastic net** (Zou and Hastie 2005) with `glmnet` (Friedman, Hastie, and

Tibshirani 2010), maximising the penalised Bernoulli likelihood. The mixing parameter α is tuned over $\{0.25, 0.5, 0.75\}$ — pure ridge and pure lasso were dropped after never winning — and λ is selected at the cross-validated minimum.

Cross-validation holds out whole fires. Observations from a single fire are strongly correlated in space and in time; splitting them across folds would report a skill the model does not have on a new fire. Folds are therefore grouped on a region-unique fire identifier, with stratified packing so that folds carry comparable numbers of positives, and K is adaptive: $K = \min(10, \text{fires with positives})$. Twenty-one of the 23 classes reach $K = 10$; two fit at 7 and 6. Out-of-fold predictions are stored per observation and are what the per-class diagnostics — calibration, omission and commission, by-fire breakdowns — are computed on.

Interactions are fitted on mean-centred factors for numerical stability and then folded back to the raw-product scale when the coefficients are exported, so that the deployed model is a plain dot product with no centring step.

3.8 Deployment in Earth Engine

The fitted models reach GEE as one small CSV of coefficients per class. Turning 23 such tables into a prediction over a whole country raises one design question, and the answer shapes the rest of the pipeline.

The obvious implementation — evaluate each class's model on its own masked subset and mosaic the results — builds 23 parallel branches of the same computation graph. Instead, **the class selection is pushed into the data**: every term becomes one image band whose per-pixel value is *that pixel's class's coefficient* for that term, written by remapping the `veg_fire` image through the 23 coefficient values. A pixel of class 7 then carries class 7's coefficient for every term, and its neighbour of class 12 carries class 12's, in the same band. Prediction becomes a single arithmetic expression, identical everywhere, and choosing the model costs nothing at evaluation time. This is also why the predictor set has to be shared across classes in the first place.

Two further properties of the deployment are worth recording. First, every multiplication renames its operands to identical band names before multiplying, rather than relying on band-matching rules, so a coefficient can never be silently paired with the wrong band. Second, a masked pixel is **masked, never zero**: cloud, no-data and non-burnable pixels contribute nothing at all, they are not observations with probability zero.

The per-observation probability is never written to an asset. Over a padded year a carta is covered by roughly 150 dates, so storing the probability per observation would mean about 150 layers per tile-year where the annual summary that follows has 16 bands — an order of magnitude more data than a product that is already on the order of a terabyte. And nothing downstream wants a single observation's probability: what carries the evidence is the *shape* of the series. Since the model is a coefficient set plus a dot product, recomputing it inside the graph is far cheaper than writing it out and reading it back. The spectral and temporal stages are therefore two concepts but one piece of code.

4 Temporal stage: burn-probability time-series metrics

4.1 Foundations

For every focal year and every carta, this stage reads the burn probability of every Landsat observation of every pixel and reduces that per-pixel series to a **16-band annual summary**: how high the probability got, how persistently it stayed there, how sharply it jumped, when, and on how many observations the answer rests. It never decides that a pixel burned. It measures the evidence that the spatial stage segments and the object model classifies.

This is the fire team's annual mosaic — except that it is a prediction, not an aggregation. Every MapBiomass initiative reduces the raw Landsat archive to one annual layer per tile and builds

everything downstream on it; for land cover that layer is the annual mosaic, a summary of *reflectance*. This layer is the fire equivalent, one level up, with a difference that decides a great deal: the model has already been applied, so what gets annualised is a *burn probability*. That buys more per byte — the vegetation-specific model, the previous-year context and the cloud masking are all resolved once, and no later stage ever touches a Landsat scene again — and it costs a dependency the plain mosaic does not have: this layer inherits the model, its training data and the land-cover collection that defines `veg_fire`, so improving any of the three means recomputing it.

Burn probability has to be smoothed before it is read as detection. A single high observation is as likely to be shadow, ash or a wet scene as fire. Collection 0 smoothed with a median over five observations. Here, vegetation types recover at very different speeds and image density varies by pixel and by year, so the stage carries **two shorter windows** — $K = 3$ and $K = 2$ — and exports both, leaving the choice between them to the next stage, per pixel.

The evidence is three things: magnitude, persistence and change, all measured in *numbers of valid observations* and never in elapsed time, because the Landsat series is irregular and its density is itself informative.

4.2 The jump, and why it is defined that way

Let $\{p_t\}$ be the series of burn probabilities of a pixel, ordered by date, and let d_t be the day number of observation t counted from 1 January of the focal year — so that padding observations from the previous year are negative, those from the next year exceed 365, and every difference is an exact whole-day count across the year boundary.

For a window of K observations, define

$$\begin{aligned} \text{minfore}_K[t] &= \min(p_t, \dots, p_{t+K-1}), \\ \text{maxback}_K[t] &= \max(p_{t-K}, \dots, p_{t-1}), \\ \text{delta}_K[t] &= \text{minfore}_K[t] - \text{maxback}_K[t]. \end{aligned} \tag{2}$$

The asymmetry between the two operators is the core of the method. Taking the **minimum** forward demands that the high signal be *sustained*: a single isolated spike does not qualify, because one low value among the next K observations collapses the floor. Taking the **maximum** backward demands that the pixel was *cleanly* low before: one high previous value is enough to cancel the jump, which is what stops a noisy low observation inside an already-burned scar from manufacturing a second detection of the same fire. delta_K is therefore the most conservative reading of a step that the series admits.

Alongside the jump, the stage records the time structure around it (Fig. 4): `jumpgap`, the number of days between the last low observation and the first high one; `prevwidth` and `postwidth`, the spans in days of the back and fore windows; and `date_post`, the day of year of the post-jump observation. `jumpgap` is strong fire evidence when *short*: a long one may be slow change — drought, dieback, harvest — rather than fire, or a fire seen through a thin image series. The two widths report how much calendar time the windows actually cover, which under sparse imagery can be most of a season.

K = 3 metrics on a burn-probability series

Same series in both panels (t^* = the observation of maximum jump). Blue = back window (3 obs), orange = fore window (3 obs).

• back window ($K = 3$) • fore window ($K = 3$) ○ other

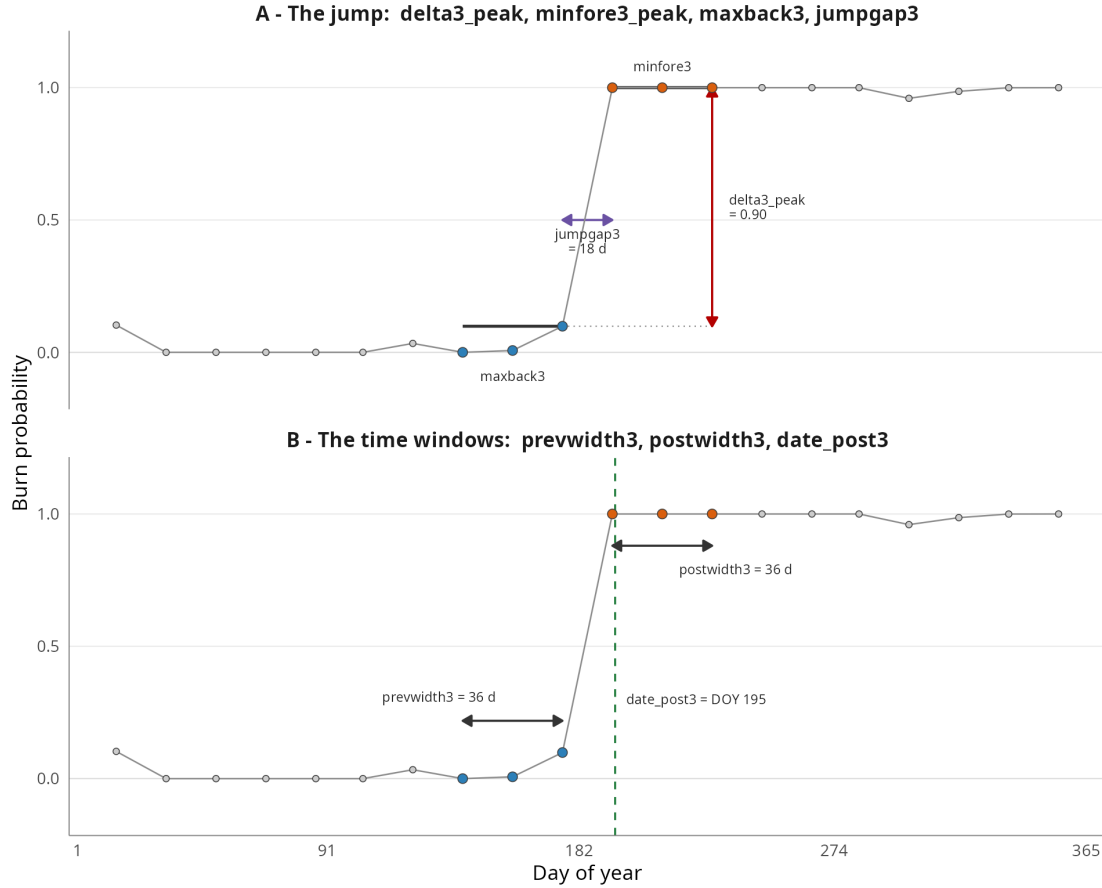


Figure 4: The $K = 3$ time-series metrics on a synthetic burn-probability series. **(A)** The jump. The black segment over the three *back* observations marks maxback_3 (their maximum); the black segment over the three *fore* observations marks minfore_3 (their minimum); the red arrow is delta_3 , the magnitude of the sustained jump, and the purple arrow is jumpgap_3 . **(B)** The time the windows span: prevwidth_3 and postwidth_3 in days, and the green dashed line at date_post_3 , the anchor from which the burn date is estimated. The $K = 2$ family is identical in construction but uses windows of two observations, which makes it more permissive and more sensitive — fewer consecutive observations are needed to confirm the jump — and therefore useful where image density is low.

4.3 The peak, and what is stored

Each delta_K is collapsed **independently** to the observation t^* at which it is maximal within the focal year, and its whole bundle of companion values is read at that observation. The two families are collapsed separately on purpose: the optimal transition observation differs by window, and tying both to one peak would mis-anchor the other.

Three further bands are not tied to any peak — the maximum raw probability of the year, and the maxima of minfore_2 and minfore_3 over the whole series, which measure sustained magnitude regardless of whether it arrived as a jump. Finally, n , the count of valid focal observations, is exported as the **sole density and quality channel**; it also carries sentinel values marking non-burnable and non-observed pixels, and it is the one band that is never masked.

Table 1 lists the 16 bands. All are encoded as signed 16-bit integers — probabilities scaled by 10^4 , days as days — which roughly halves the size of the asset. Missing data is never given a probability: a pixel with no valid observations comes back with $n = 0$ and everything else masked. A pixel whose series is too short for a given window has *that window's* bands masked, which is a quality flag rather than an error.

Table 1: The 16 exported bands of the temporal stage. Each jump metric exists in two families, $K = 3$ (conservative) and $K = 2$ (permissive), collapsed at their own t^* .

Concept	Band(s)	What it measures
Change	delta3_peak , delta2_peak	the sustained jump — the central fire indicator
Persistence	minfore3_peak , minfore2_peak	the probability floor just after the jump
Sharpness	jumpgap3 , jumpgap2	days between the last low and the first high obs.
Window span	prevwidth3 , prevwidth2	days spanned by the back window
	postwidth3 , postwidth2	days spanned by the fore window
Date	date_post3 , date_post2	day of year of the post-jump observation
Magnitude	pmax1	maximum raw probability of the year
	pmax2 , pmax3	maximum sustained probability of the year
Quality	n	count of valid focal observations

Note that maxback_K is not stored: it is exactly $\text{minfore}_K - \text{delta}_K$ and is recovered whenever it is needed. Note also that date_post is *not* the fire date. It is the day of the post-jump observation, and the burn date used downstream is the mid-point between the last low and the first high observation, $\text{date_post}_2 - \text{jumpgap}_2/2$.

4.4 Crossing the year boundary

Window metrics starve at the extremes of a series — maxback_3 needs three observations behind it, minfore_3 two ahead — and in a fire-prone country the start and the end of the calendar year are in the middle of the burning season. The focal year therefore **borrows three observations from the previous year and two from the next**: asymmetric on purpose, because symmetric padding of two would under-supply the back window at the first focal observation.

The neighbouring observations cannot simply be taken from the adjacent scenes, since the literal nearest scene may be cloud-masked at this pixel. Burn probability is instead computed for a margin of a few months on each side and the three latest previous and two earliest next *valid* observations are kept per pixel; the margin is widened for 1999 and 2000, where the archive is thin. Padded observations use the focal year's previous-year context, which is strictly the wrong year for them but is the better approximation over three-plus-two observations.

Crucially, **the search for t^* is always restricted to focal observations**. Padding is only ever *context*, so a burn can never be detected twice in two adjacent years — which means that, unlike the pilot, this collection has no cross-year duplicates to remove later.

4.5 Cost

This is the most expensive artifact in the collection, in compute and in storage: a complete year is roughly 55 GiB over the 248 cartas, and the 27 years are on the order of 1.4 TiB. Per-account task parallelism in GEE is limited, so a whole-country year cannot be produced by one person in reasonable time; production ran as a distributed, multi-account export, one contributor per year, with an idempotent launcher that skips any tile which already has a finished asset or a running task.

Two consequences follow for anyone planning a change. First, the optimisation levers are all *product-level* — number of terms, window length, band set — and each one changes the model or the output; there are no late tweaks to be had, because profiling puts the entire array and time-series machinery below 1 % of per-tile cost, with the expense sitting in the model arithmetic and the cloud masking over roughly 150 scenes. Second, **re-running this layer is a collection-level decision**, not a step-level one.

5 Spatial stage I: segmentation into scar objects

5.1 Foundations

Seeds and candidates are how this stage avoids deciding early. A single probability cut would force one threshold to be both sensitive and specific. Instead a *strict* cut defines where a scar certainly started — the **seeds** — and a *permissive* cut defines where one could plausibly extend — the **candidates** — and region growing resolves the two: a candidate joins a scar only if it is connected to a seed, and a candidate island with no seed in it is dropped altogether (Fig. 5).

This exploits a favourable trade-off between the two error types. Seeds have high omission and low commission error; candidates the opposite. A pixel whose own evidence is weak is accepted when it belongs to a patch whose evidence is strong, and rejected when it stands alone. Region growing has been used this way in burned-area mapping for decades, exactly because fire is spatially contagious (Adams and Bischof 1994; Haralick and Shapiro 1985; Fraser, Li, and Cihlar 2000; Giglio et al. 2009; Hawbaker et al. 2020; Long et al. 2019).

The stage errs deliberately towards **recall**: precision is recovered at the object classifier of Section 6.4, which can see the patch's shape, size and seed density — things no pixel can report.

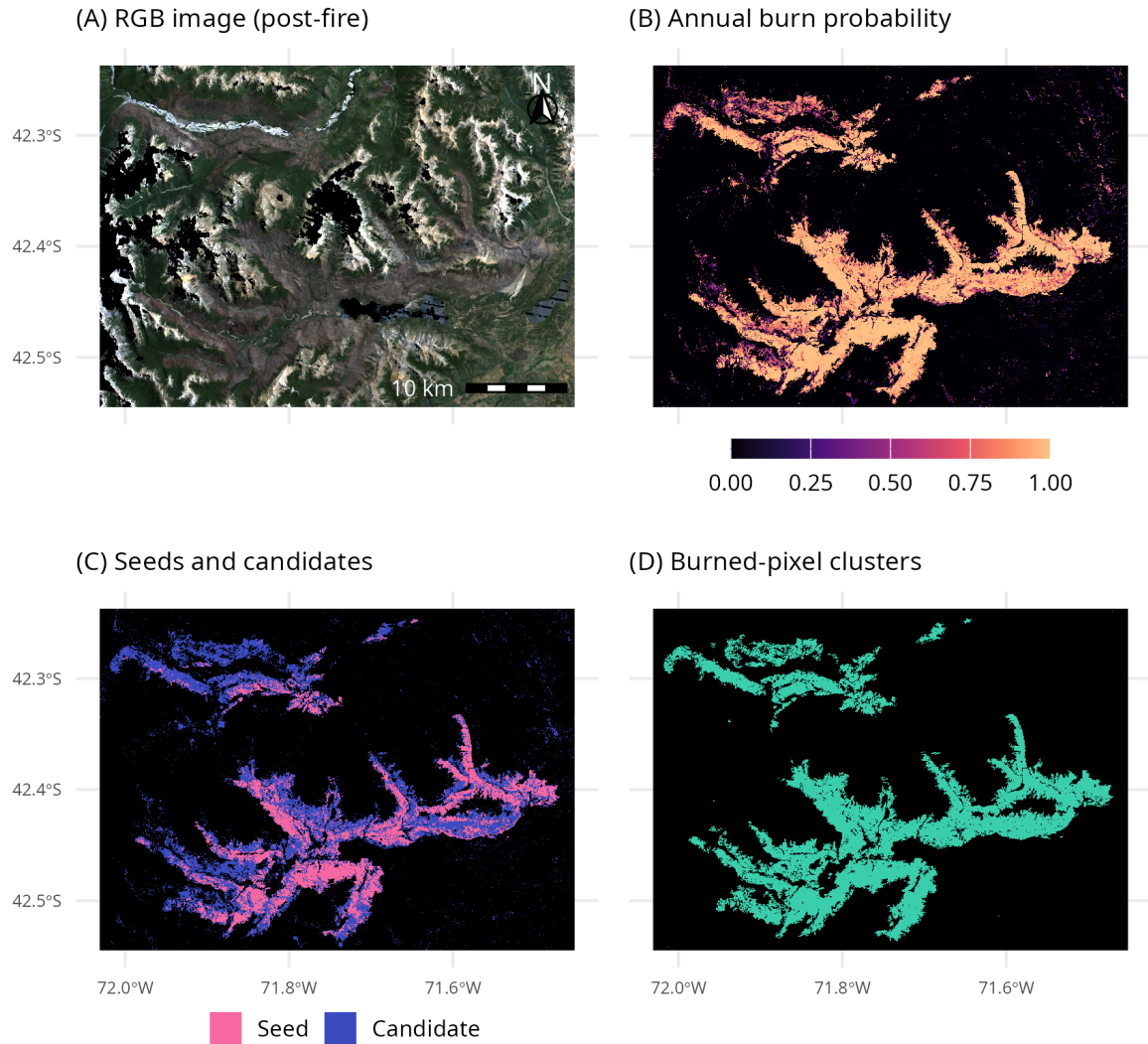


Figure 5: The region-growing step, illustrated on two fires that burned in 2015 in Chubut province (Río Turbio to the north, Cholila to the south). **(A)** RGB composite of the following summer, for reference only; black areas are masked by the quality filters. **(B)** The per-pixel evidence of burning. **(C)** Seeds and candidates, obtained by applying a strict and a permissive cut to that evidence. **(D)** The output of the region growing: the candidate pixels connected to at least one seed. *The panels were produced for the Collection 0 pilot, whose per-pixel evidence was an annual burn probability; in Collection 1 the same two cuts are applied to the jump metrics of Section 4. The logic of the step is unchanged.*

5.2 The fire year

A fire is an object in space *and* in time, and this stage buys the time axis with a calendar rather than with an algorithm. Segmenting calendar years independently splits every scar that straddles 31 December and duplicates it across two years. The obvious repairs — a temporal firebreak inside the segmentation, a backward gap-fill from the previous year — were built first and did not work, because the per-pixel dates are too noisy to serve as a barrier. The problem disappears entirely if the year boundary is placed where almost nothing is burning.

The **fire year** therefore runs from **1 May of year Y_1 to 30 April of year $Y_1 + 1$** and is named after its starting year. May is the country-wide trough of fire activity — in MODIS and VIIRS active fires and in our own burn dates — so no region's season is cut in half: the summer burners of the south and centre and the winter-spring season of the north both fall inside a single May-to-April year. One national boundary is a compromise; Patagonia would ideally break in June or July. Defining a separate fire year per region was tried and abandoned, because processing the whole country in one pass brought far more benefit than the optimum boundary would have.

Since the annual metrics cover calendar years 1999–2025, the fire years mapped are 1998–2025, with two **partial** edges: fire year 1998 holds only the January–April 1999 tail, and fire year 2025 only the May–December 2025 head. Both are flagged as partial in the asset metadata, and anything that averages across fire years has to know.

5.3 Seeds, candidates and dates

A fire year spans two calendar-year metric images, and each is thresholded before they are combined.

Candidates are pixels whose delta_2 exceeds a permissive cut — always the $K = 2$ family, because it gives the broadest footprint. **Seeds** are pixels whose delta_K exceeds a strict cut, with K **chosen per pixel** from its vegetation class and its observation count n : a pixel whose series is dense enough for its class is judged on the conservative $K = 3$ metric, and a pixel in a thin part of the archive on the permissive $K = 2$ one. A temporal-gap gate also applies, rejecting seeds whose jumpgap exceeds an n -adaptive ceiling — that is, rejecting changes too slow to be fire, which is precisely the drought signature that forced hand-drawn masks in the pilot. Cuts are set per vegetation class, with global defaults, and non-vegetated classes receive an unreachable cut so that they can never fire.

Each pixel's **burn date** is taken from the $K = 2$ family as $\text{date_post}_2 - \text{jumpgap}_2/2$, converted to an absolute day count since a fixed epoch, so that every later date comparison is safe across year boundaries.

The two calendar images are then **filtered by date and combined**. Only pixels whose burn date falls inside $[1 \text{ May } Y_1, 1 \text{ May } Y_1+1)$ are kept: the Y_1 image contributes May–December, the Y_1+1 image January–April, and a detection dated before May of Y_1 belongs to the previous fire year and is dropped. Where both images claim a pixel, they are combined by rank — seed beats candidate beats nothing — and the date follows the image that won. This is what turns two calendar layers into one non-calendar fire year.

5.4 Slow post-fire dieback in Andean Patagonia

Andean Patagonian forest dies *slowly* after fire: part of a real scar crosses the change thresholds only in the *following* fire year, when the canopy finally collapses. For forest and shrubland pixels west of a longitude cut, a pixel that is seed or candidate in the next calendar image with a mid-winter to spring date is added to the focal year as a **candidate**, flagged with its own code — never as a seed, because dieback must be able to *extend* a detected scar but never to *create* one. Since padding survives region growing only where it connects to a genuine focal seed, it cannot manufacture a fire. These pixels carry a next-year date, so they are excluded from every date statistic downstream and inherit their parent object's date instead.

5.5 Supervised region growing

The region growing itself uses SNIC (Simple Non-Iterative Clustering, Achanta and Süsstrunk 2017), as implemented in GEE. SNIC is an unsupervised segmentation method, and its most common use is to partition a wall-to-wall image from uniformly placed seeds and spectral features. Here it is used in a far more guided way: we supply the seeds and we supply the footprint through which they may grow, so what would ordinarily be a segmentation becomes a classification — SNIC is the mechanism, and the seed/candidate definition is the model.

Two details matter. Seed clumps below a small size are **demoted to candidates** before growing: they no longer seed, but they remain in the footprint so that a genuine cluster can still grow through them. And the growing runs on the **whole country in one pass**, not region by region: SNIC's memory footprint depends on its internal tile size and neighbourhood buffer, not on the extent of the export, so a national run costs what a regional one costs, and a large neighbourhood heals the internal seams. Only the *mask* of the result is kept — the cluster identifiers are discarded, because the objects are relabelled globally in the next stage.

The output is one image per fire year with a single band distinguishing candidate, seed and dieback padding, masked everywhere else. Burned pixels are 0.3–1 % of the country, which is what makes the whole national layer small enough to download.

6 Spatial stage II: fire objects and the object classifier

6.1 From pixels to objects

This stage materialises the fire patch so that it can be judged as a patch. A patch has properties no pixel has — shape, size, vegetation composition, a spread of burn dates — and those are what separate a real scar from a field of spectrally similar noise.

Objects are the connected components of the burned pixels, labelled **one fire year at a time, whole-country and untiled**, so that an object is global within its year and a scar is never split by a processing seam. Labelling uses a union-find structure (Tarjan 1975), which needs only one integer per burned cell; the country's 30 m lattice holds 9.16 billion cells while a heavy fire year holds around 116 million burned ones, three orders of magnitude fewer, and every stage of this part of the pipeline is sized by the latter.

Connectivity is wider than eight-neighbour adjacency, and selectively so. A real scar frequently breaks into fragments one or two pixels apart, and gluing them recovers one fire instead of five. The rule is equivalent to dilating the burned mask by one pixel, labelling, and dropping the halo — but it is implemented as a wider union window, which avoids ever materialising the dilation. Two burned cells are joined if their Chebyshev distance is at most 3 when both have *enlarged context*, at most 2 when only one does, and at most 1 — plain eight-connectivity — when neither does. A cell is denied enlarged context when it is agriculture, pasture or temperate grassland, where bridging distinct burned fields would inflate commission error, or when it is a dieback pixel, which may only extend a scar. Everything else — forests, shrublands, the arid grasslands — keeps it, because in sparse fuels bridging recovers a single real scar.

Each labelled object is then vectorised into one, possibly multi-part, polygon. Disconnected fragments joined by the bridging rule dissolve into a single multipolygon, so the bridge is preserved natively in the geometry.

6.2 Object metrics

The metric set is deliberately narrow — only what the classifier uses — and is computed directly over the burned cells rather than over the polygons, which is both faster and exact for area.

- **Seed fraction:** the share of the object's pixels that are seeds. This is the single most discriminating metric for fire against noise: real scars are densely seeded, noise is not.

- **Neighbourhood density:** the mean, over the object's pixels, of the burned fraction within square windows of one, two and three pixels' radius. Sparse, threadlike objects score low; solid scars score high.
- **Size:** area in hectares, and pixel count.
- **Dates:** the median, minimum and maximum burn date of the object's pixels, and the calendar year that most of its pixels fall in.
- **Vegetation composition:** the fraction of the object's pixels in each `veg_fire` class.
- **Shape:** perimeter, convexity (area over convex-hull area), bounding-box fill and elongation, circularity $4\pi A/P^2$ and shape index $P/(2\sqrt{\pi A})$.

Area is measured on the ellipsoid, which means a pixel is not 0.09 ha: at these latitudes one Landsat pixel covers about 831 m² at 22°S and 517 m² at 55°S. A size class is therefore a *range* of pixel counts, and the same 15-pixel object changes size class between the north of the country and Patagonia.

6.3 Object labels

The classifier is fitted on **about five thousand fire / non-fire labels collected by hand**. Collectors work in the GEE Code Editor on the seed-and-candidate layer itself — not on uploaded polygons — because that layer already shows, per pixel, candidate against seed on exactly the clusters the region growing kept, which makes shape and seed density, the most informative signal, fully visible without any vector data. Each collaborator exports a single asset; drawing-layer names carry the class and the fire year. Labels are then matched to *objects*, not to pixels, and the object inherits the label.

Cleaning the label-object pairs — dropping labels that hit no object, objects labelled both ways, and duplicate labels on an already-labelled object — yields a fitting set of **5,255 objects**, 2,788 fire and 2,467 non-fire.

The labelled sample is not a random sample of objects, and this single fact conditions every number the model produces. Labels were collected where fires were known to be, so per-year prevalence in the fitting set runs from 0.00 to 1.00, seven fire years have no labels at all, and small objects are strongly under-sampled. The model's *ranking* and its *uncertainty* are the product, not a calibrated probability; the deployed decision thresholds are a lower bound; and the cross-validation design has to be chosen with this sampling in mind. Fixing it is a *collection* task, not a modelling one: a randomly sampled set of small-object labels.

6.4 The classifier

A classifier, not a threshold filter. Collection 0 decided fire with a hand-built rule over size, shape and neighbourhood density. Reproduced verbatim against our labels, that filter scores an accuracy of 0.62 and a sensitivity of 0.50, against 0.81 and 0.85 for the model at the deployed cuts — and it fails worst exactly where the objects are, discarding 81 % of the real fires in the 1–50 ha band, which holds 83 % of all objects. Two of its ideas survive: a hard minimum-size cut, and reasoning stratified by size.

The model is a probit BART (Chipman, George, and McCulloch 2010; Albert and Chib 1993), fitted locally in R with `stochtree` (`stochtree` developers 2025). The choice follows from what five thousand labels allow. There is no honest way to tune a boosted ensemble's hyperparameters on a set this small, whereas BART's defaults are calibrated regularisation priors rather than placeholders. More importantly, the posterior yields a **per-object interval** on the probability — both an uncertainty statement that can be published and the targeting signal for a second round of label collection. The fit takes about ninety seconds; scoring the whole object set is done fire year by fire year in blocks, so that peak memory is one block however many objects exist.

The model is applied to **all 28 fire years and 1,689,419 objects**.

6.5 Predictors, and why none may name the year

Twenty predictors are used: fifteen size, shape, density, seed and date-structure metrics, plus **five aggregated vegetation fractions** — agriculture, flooded grassland, pasture, temperate grassland and woody. The aggregation replaced the 23 raw class fractions, which were 58 % of the design matrix and consumed the model’s split budget without adding skill; membership of each group is derived from the remap table *by name*, so a remap change follows through automatically. The five groups are deliberately not a partition and do not sum to one.

No predictor may name the year or act as a proxy for it, and the rule has already cost three candidates. Per-year label prevalence in the fitting set is an artifact of where people drew, so a model given the year learns that sampling pattern and applies it to every object of that year. The fire year and the calendar year were removed for leaking the labels outright — out of fold, the prediction for a year *was* its label prevalence — and the mean observation count was removed as a *soft era proxy*: observation density rises as sensors come online, which lets an improving archive masquerade as a rising fire regime. All three keep their roles as product attributes; none of them is a predictor.

The same rule is why season enters as $\sin$ and $\cos$ of the day of year rather than as the day itself — season is legitimate information, but it must be **circular**, since the fire season straddles the turn of the year — and why no absolute time coordinate is a predictor, burn dates included. Note that spatially blocked cross-validation cannot detect this class of leak, because it blocks space and leaves the year on both sides of every split; a standing per-year check is run instead.

6.6 The decision threshold

A cut of 0.5 is wrong, and the right cut rises with object size. Thresholds are swept on *out-of-fold* probabilities — never in-sample, or the cut would be chosen against answers the model has already seen — and selected by Youden’s J (Youden 1950), which is the only criterion here that does not move with prevalence.

Table 2: Deployed decision thresholds by size band, selected on out-of-fold probabilities by Youden’s J . The <1 ha row is shown for completeness but is not leaned on: there, a hard size cut rather than a threshold is the right tool.

Size band	labels	prevalence	cut	J at the cut	J at 0.5
< 1 ha	114	0.254	0.250	0.826	0.643
1–50 ha	3217	0.468	0.202	0.600	0.492
50–300 ha	1192	0.576	0.436	0.645	0.631
≥ 300 ha	732	0.773	0.690	0.706	0.662

The bootstrap intervals on the three deployed cuts are close to disjoint, so the differences are signal rather than noise: the model is far more confident on large objects, and a single threshold would be simultaneously too high for small objects and too low for large ones. The gain lands where the error was — sensitivity in the 1–50 ha band rises from 0.596 to 0.837.

Two qualifications. First, **the threshold governs object counts, not the area headline**: on a heavy year the band cuts call some 64,000 objects fire against 50,500 at a flat 0.5, for an increase of well under 2 % in burned area, because area is dominated by large objects the model is confident about. Second, **the cuts are a lower bound**. J is prevalence-invariant as a *measure*, but the cut it selects is optimal for the prevalence of the set it was selected on, and that set is not the population (Section 6.3).

Where a human label exists for an object, it **overrides** the model’s call. The published call is therefore the collected label where there is one and the model’s call otherwise, and the three columns are kept separate in the product so that the provenance of every call is visible.

6.7 Performance

Cross-validation blocks space into 0.5° cells assigned to five folds. This removes the adjacency leak — objects from one drawn polygon are neighbours — while keeping every region in every fold, which is the deployment condition. Leave-one-region-out was run for comparison and is the harshest possible test, but it is not the condition the model is used in: every region does have labels in production, and held-out prevalence swings from 0.10 in Patagonia to 0.83 in the Pampas, so a fold's model would be trained on a class mix it is never scored on. **The fold design decides the answer**, and the choice has to be argued rather than defaulted.

Table 3: Spatially blocked cross-validated performance, pooled over folds, at a flat 0.5 cut and at the deployed per-band cuts. AUC is 0.891 pooled; the observation-weighted *within-year* AUC is 0.845 over the eighteen years carrying both classes — that is the figure to watch for leakage.

	at 0.5	at the deployed cuts
Accuracy	0.790	0.814
Sensitivity	0.717	0.845
Specificity	0.872	0.780

Performance is not uniform across sizes. Above 300 ha the model is nearly clean (AUC 0.923); the weak band is **1–50 ha** (AUC 0.871), which is also where 83 % of the objects are, and where at a flat 0.5 cut the model would miss two fires in five. That band is 61 % of the labels but 83 % of the population — the same under-sampling that the threshold caveat and the posterior widths both report.

Variable importance was assessed with four different measures, because the predictors are strongly correlated — area with pixel count with perimeter, the three nested neighbourhood windows — and each measure mishandles correlation in its own way. Partial-dependence curves were avoided for the same reason and **accumulated local effects** (Apley and J. Zhu 2020) used instead: a partial dependence plot averages over combinations that do not exist, such as a one-pixel object with a ten kilometre perimeter, and invents effects there. **Two predictors carry the model**: the temperate-grassland fraction and the seed fraction. That is the intended story — real scars are densely seeded, and the fuel type decides how a burned patch looks. The size and shape block is nearly inert, and pixel count is last of the twenty on every measure. Reassuringly, no predictor shows the signature that exposed the year leak: a large split share concentrated near the root of the trees with an effect that tracks the calendar.

6.8 The minimum mapped size

Scoring the full population answers the question the minimum-size decision was supposed to rest on — *is the model measurably less able to classify small objects?* — and the answer is: not distinctively. Posterior uncertainty does fall monotonically with size, with mean interval width going from 0.41 below half a hectare to 0.13 above a thousand hectares, but the model is **unsure everywhere**: the global mean width is 0.32 and 31 % of all objects have an interval that straddles their own decision cut.

Two conclusions follow. **The honest argument for a one-hectare minimum is cost and benefit, not uncertainty**: it discards 3.4 % of the objects for 0.044 % of the area. And the interval width is reporting *covariate shift* — BART widens its posterior where it has no data, and 31 % undecided is that message about five thousand labels against 1.69 million objects. The one-hectare cut is applied as a hard rule, which also satisfies, more strictly, the solitary-pixel filter of the network's common post-processing, since one hectare is about eleven pixels.

7 Products

Steps one to six are the algorithm; what follows is delivery. The fire-year objects are re-partitioned into **calendar years**, with the year and the month assigned **per pixel** from that pixel's own burn date rather than per object, and the published layers are derived from those pixels. Two of the network's post-processing stages — the non-burnable land-cover mask and the solitary-pixel filter — are already embedded further upstream in this algorithm and more strictly than the specification asks, so running them again would be a no-op at best.

7.1 The six subproducts

The published products are the ones the MapBiomias Fuego network defines. They are not Argentina's design and they are not justified here: every participating country publishes the same set, in the same shape and with the same pixel encodings, so that the layers are comparable across the network and readable by the same platform. Argentina delivers all six — *annual*, *monthly*, *accumulated*, *frequency*, *year of last fire* and *scar size*; some countries deliver only the first four.

The six become twelve assets: four have a *coverage* variant crossing the burned pixels with the published MapBiomias Argentina land cover of the same year, and the scar-size product comes with the scar identifier and the scar area it derives from. Every asset is a **single multiband image with one band per year**, not one image per year — 27 bands (1999–2025) for the annual, monthly and year-of-last-fire products, 53 for the frequency and accumulated ones, one per two-sided window (Table 4).

Table 4: The published raster subproducts. M is the month of burn (1–12, masked elsewhere) and L the land-cover class of the same calendar year. Asset names follow the network convention `mapbiomas_argentina_fire_collection1_<subproduct>_v<N>`.

Subproduct	Band name	What a pixel holds
<code>annual_burned</code>	<code>burned_area_<year></code>	1 if burned that year
<code>annual_burned_coverage</code>	<code>burned_coverage_<year></code>	L where burned
<code>monthly_burned</code>	<code>burned_monthly_<year></code>	M , the month of burn
<code>monthly_burned_coverage</code>	<code>burned_coverage_<year></code>	$M \times 100 + L$
<code>frequency_burned</code>	<code>fire_frequency_<y1>_<y2></code>	number of years burned in the window
<code>frequency_burned_coverage</code>	<code>fire_frequency_<y1>_<y2></code>	$\text{frequency} \times 100 + L$
<code>accumulated_burned</code>	<code>fire_accumulated_<y1>_<y2></code>	1 if burned at least once in the window
<code>accumulated_burned_coverage</code>	<code>fire_accumulated_<y1>_<y2></code>	L where burned at least once
<code>year_last_fire</code>	<code>classification_<year+1></code>	calendar year of the most recent fire
<code>annual_burned_id</code>	<code>scar_id_<year></code>	identifier of the scar the pixel belongs to
<code>annual_burned_area_ha</code>	<code>scar_area_ha_<year></code>	area in hectares of that scar
<code>annual_burned_scar_size_range</code>	<code>scar_area_ha_<year></code>	size class of that scar, 1–8

A **scar** is a connected group of burned pixels within one calendar year, labelled with plain eight-connectivity — the network's definition, and deliberately not the wider connectivity the fire objects of Section 6 use. The eight size classes are the platform's published legend: <10, 10–250, 250–500, 500–5,000, 5,000–10,000, 10,000–50,000, 50,000–100,000 and $\geq 100,000$ ha. Argentina populates all eight; the largest scar in the series is 219,410 ha, in 2003. The scars also exist as vectors, one FeatureCollection per calendar year.

Two reading rules follow from how the pixels are built. **The month of burn is measured, not inferred**: it comes from the pixel's own burn date rather than from the date of a minimum-NBR composite, which is how most of the network derives it. And **a fire that straddles 31 December is two scars**, one in each calendar year — the price of assigning the date per pixel, and what makes the annual, monthly and scar-size layers agree pixel for pixel.

7.2 The fire-object vector database

Alongside the six, and not one of them, Argentina publishes the **fire objects themselves** as a single vector layer: `burned_area_polygons_v2`, **1,012,645 objects covering 63.33 Mha**, every mapped fire of all 28 fire years in one FeatureCollection. It is the output of Section 6 under the published selection, with nothing recomputed and no geometry touched.

Each polygon carries ten properties: `oid`, the stable object identifier that joins back to the object database and its twenty metrics; `fire_year`, the non-calendar mapping year; `calendar_year`; `area_ha`; the three burn dates `date_med`, `date_min` and `date_max`, written as YYYY-MM-DD; `p_mean` and `p_width`, the object classifier's posterior mean fire probability and the width of its credible interval; and `seed_mean`. Every feature is stamped with `system:time_start` from `date_med`, so the collection answers `filterDate()`; `system:time_end` is deliberately not set, so that one fire falls in exactly one time bucket and filtered areas stay summable.

Three things a user of this layer must know.

- **It is a fire-year layer, and the rasters are calendar-year layers.** `calendar_year` is the *majority* year of the object's pixels, while every raster above assigns year and month per pixel. A fire straddling 31 December is split across two years in the rasters and lands whole in one year here. Neither is wrong, but cross-tabulating the two without knowing this produces apparent "missing" area.
- **Fire year 1998 exists here and in no published raster.** The calendar series starts in 1999, so the November–December 1998 tail — 3,845 polygons, about 76 kha — is only in this layer.
- **Area must be summed per object, not per row.** One very large fire year 2000 object exceeded Earth Engine's vertex limit on ingest and is stored as four features that each repeat the whole object's attributes. Summing `area_ha` row by row overstates the layer by 5.1 Mha.

7.3 Where the products live

The assets above are produced in the project's own Earth Engine folder:

```
projects/mapbiomas-argentina/assets/FIRE/COLLECTION-1/FINAL_PRODUCTS/
```

That is **not** where a user should look for them — it is the production tree, its contents are versioned and rebuilt, and access to it is restricted to the team.

The six subproducts reach the public through the MapBiomass Fuego network: they are copied, by an explicit subproduct list, into the network's public asset collection, and from there they are rendered on the MapBiomass Argentina platform, where they can be browsed by year, month, frequency and territory and downloaded as statistics. The entry point for the general public is <https://argentina.mapbiomas.org/>, and the fire section of the platform is the interactive view of exactly these layers.

The fire-object vector database is different: it is **Argentina's own layer**, not one of the network's six, so it is not swept into the public copy. We share it directly, as a link to the asset above.

[To be completed before release.] The exact public locations — the platform URL of the fire section, the public asset path of the network copy, the download page for the statistics, and the form in which the vector database is shared — have to be filled in here once the launch configuration is fixed with the network and with the MapBiomass Argentina coordination.

7.4 Using the products in Earth Engine

The quickest way to see the products is the Earth Engine Code Editor. The snapshot below loads the annual and monthly layers for one year, the frequency layer for the whole series, and the fire-object polygons, with the visualisations the platform uses:

<https://code.earthengine.google.com/SHOT-1D>

The script it opens is essentially the following, which can also be pasted into a blank Code Editor tab:

```
var ROOT = 'projects/mapbiomas-argentina/assets/FIRE/COLLECTION-1/FINAL_PRODUCTS/';
var NAME = 'mapbiomas_argentina_fire_collection1_';
var year = 2020;

// Annual burned area: one band per year, 1 = burned.
var annual = ee.Image(ROOT + NAME + 'annual_burned_v2')
    .select('burned_area_' + year).selfMask();
Map.addLayer(annual, {palette: ['#ff0000']}, 'burned ' + year);

// Month of burn: 1-12.
var monthly = ee.Image(ROOT + NAME + 'monthly_burned_v2')
    .select('burned_monthly_' + year).selfMask();
Map.addLayer(monthly, {min: 1, max: 12, palette: ['#2c7bb6', '#ffffbf', '#d7191c']},
    'month of burn ' + year);

// Fire frequency over the whole series: number of years burned.
var freq = ee.Image(ROOT + NAME + 'frequency_burned_v2')
    .select('fire_frequency_1999_2025');
Map.addLayer(freq, {min: 1, max: 10}, 'fire frequency');

// The fire-object vector database: one feature per mapped fire.
var fires = ee.FeatureCollection(ROOT + 'burned_area_polygons_v2')
    .filterDate(year + '-01-01', (year + 1) + '-01-01');
var style = {color: 'white', fillColor: '00000000', width: 1};
Map.addLayer(fires.style(style), {}, 'fire objects ' + year);

Map.setCenter(-64, -34, 5);
```

Note the two patterns every consumer needs: a subproduct is one image and a year is one **band** of it, and the polygon layer is filtered by date rather than by a year property, because each fire carries a single timestamp taken from its median burn date.

8 Reproducibility

The code and the documentation needed to reproduce this workflow are hosted on GitHub. The repository is organised as one directory per collection; Collection 1 lives in `collection-01/`, with the numbered pipeline scripts in `workflow/`, the configuration tables that every stage reads in `config/`, the fitted coefficients in `models/`, and the analysis notebooks in `notebooks/`. The per-step technical notes in `collection-01/docs` are the intended entry point for anyone reproducing or adapting the method: each one states what its stage measures, lists its inputs and outputs, records the decisions and their reasons, and warns about the failure modes that were actually encountered. The Earth Engine JavaScript used for interactive threshold tuning and for label collection lives in a separate Code Editor repository, referenced from those notes.

9 Limitations and known challenges

9.1 Where the algorithm struggles

Productive, actively managed landscapes are the hardest case. In agricultural regions the region-growing step produces poor clusters: harvest patches enter the candidate footprint and are

occasionally merged into a genuine scar. No spatial strategy can repair this, because the ambiguity is already present in the per-observation probability — a harvested field and a burned field look alike to a single-date spectral model. Improving fire mapping in croplands and sown pastures requires better *spectral* models, which in turn requires collecting training data specifically on those confusions.

The object classifier was conceived from the pilot's experience, where many false positives were easily separable from fire by shape and seed density: dendritic, sparse objects associated with drought and volcanic ash. It does that job well, and it also removes clearing-type disturbances — roads, urbanisation — that the pilot confounded with fire. But in regions where the false positives are compact and densely seeded, it helps much less. It is a filter on *patch geometry and density*, and it cannot recover information the spectral stage did not capture.

Understorey fires in the most productive systems remain largely undetected, as they were in the pilot: a fire that burns beneath an intact canopy leaves little signal in a 30 m top-of-canopy reflectance series.

9.2 Limitations inherited from the data

The object classifier's labelled sample is the binding limitation of the whole spatial stage (Section 6.3). Until a randomly sampled set of labels exists, the deployed thresholds must be treated as a lower bound and the per-object probability as a ranking rather than as a calibrated probability.

Observation density varies enormously through the series, and the algorithm handles it explicitly — an adaptive window choice, an adaptive gap gate, and an exported quality band — but handling is not equivalence. Detections in 1999–2001, when only two sensors were flying and one of them had just launched, rest on fewer observations than detections in 2015. A residual increasing trend in the fire-object rate survives the removal of the era-proxy predictor; it is **unattributed rather than demonstrated to be clean**, and it should not be published as a fire-regime finding without an independent record to check it against.

The edge fire years are partial by construction (Section 5.2): 1998 holds only four months and 2025 only eight. Completing them requires extending the temporal stage back to May 1998 and forward through 2026.

The land cover used to select models is frozen. The `veg_fire` classification comes from a specific MapBiomas LULC collection, deliberately held fixed, because it decides which model judges each pixel and therefore the entire downstream candidate set. A newer land-cover collection reassigns that wherever a class changed, so the layer cannot be half-migrated: either the whole collection moves or none of it does.

9.3 Limitations of shape

A fire that crosses 31 December becomes two scars in the published calendar-year products, because year and month are assigned per pixel. Within the algorithm the fire is one object in one fire year; the split is a property of the delivery format, and the fire-object layer preserves the undivided event.

The minimum mapped unit is one hectare. Smaller objects are scored like everything else but are not published; they are 3.4 % of the objects and 0.044 % of the area.

10 Priority lines of improvement

These are listed in what we judge to be decreasing order of effect on mapping accuracy. They do not consider incorporating sensors other than Landsat, since for the first collections the priority is to refine the mapping without compromising the temporal extent of the product.

Training data that represents the false positives. As in the pilot, the largest available gain is data on the processes that mimic fire — harvest, drought, flooding, logging, insect defoliation, ash deposition — collected explicitly as negatives, region by region. The Collection 1 training set already contains some of these, and they demonstrably help; there are nowhere near enough of them. Because these processes are spectrally similar to burning, the *relative* abundance of each class in the training set matters as much as the absolute number of samples: the less separable the classes, the more the prior implied by the sampling drives the fitted probability.

A randomly sampled set of object labels. The second-largest gain costs the least effort. A few thousand labels drawn at random from the object population — rather than from where fires were known — would convert the object classifier’s output from a ranking into a calibrated probability, put the size-band thresholds on a defensible footing, and let the population fire rate be estimated rather than assumed. Objects with wide posterior intervals are the natural targets for a second round.

Better use of the temporal metrics in the segmentation. The sixteen exported bands were fixed before the segmentation stage existed, and some of them are never read. The interesting question is not only which to drop but **which the segmentation should have been using**: the seed and candidate cuts currently rest on the jump metrics and the observation count, leaving the sharpness and the window-span bands — the ones that best separate abrupt change from gradual change — almost unused. This is the cheapest available improvement in commission error, because it requires no re-export.

Splitting the vegetation remap. Several `veg_fire` classes are currently merged across regions for want of data rather than on ecological grounds (Section 3.4). With more collected fires, those groups can be split and fitted on their own merits, which directly improves the per-class fit where it is weakest.

The spectral model itself. The logistic regression can be extended to a categorical regression predicting the probability of several classes at once — unburned, burned, cloud, water, ash, clearing, drought — instead of a binary one; or simple auxiliary models can be multiplied into it, for instance a brightness-based probability of cloud, snow or ash. Both remain compatible with the constraint that the model has to run inside GEE over the entire archive. This is also where richer classifiers would pay off most, if a deployment path for them existed.

Provenance stamping. A practical, cheap improvement: every exported annual-metrics asset records its year and its tile but *not* which coefficient set produced it. Because the coefficients enter through a per-pixel remap, a model improvement affecting one vegetation class changes nothing in tiles where that class is absent, so a partial re-export is technically sound — but it would be of undocumented provenance. Stamping the deployed model version on every asset costs nothing and is what makes a partial re-run auditable.

11 Considerations for Collection 2

Re-running the temporal stage is the single largest commitment a collection can make — on the order of a terabyte and weeks of distributed export — so the decisions that force it belong on the table early, with time to think, rather than in the run-up to a launch. Three of them are already identified: whether to recompute against a newer land-cover collection, which reassigns `veg_fire` everywhere a class changed and therefore cannot be done by halves; whether a model improvement may be applied to part of the country, which is technically possible and currently blocked only by the provenance-stamping gap above; and which of the sixteen bands to keep.

Two metrics should simply stop being computed. The mean observation count per object is not a predictor and carrying it invites the era-proxy mistake; the observation count belongs where it is already used and normalised, inside the seed definition. And pixel count should give way entirely to area, since the pixel scale is latitude-dependent and the model confirms that the count carries nothing the area does not.

Beyond that, the two open fronts are the ones named above: training data on the confounding processes, and a randomly sampled set of object labels. Both are collection effort rather than algorithmic development, and both are the kind of work whose payoff is invisible until it is done.

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